Started on	Monday, 28 September 2026, 9:29 PM
State	Finished
Completed on	Monday, 28 September 2026, 9:48 PM
Time taken	18 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int findFloor(int a[],int low,int high,int x)
{
    if(low > high)
        return-1;

    if(x>=a[high])
        return a[high];

    if(x<a[low])
        return-1;

    int mid = (low+high)/2;

    if(a[mid]==x)
```

```
        return a[mid];

    if(a[mid] < x)
    {
        int result = findFloor(a,mid +1,high,x);

        if(result == -1)
            return a[mid];
        return result;
    }
    return findFloor(a,low,mid-1,x);
}

int main()
{
    int n,x;
    scanf("%d",&n);
    int a[n];

    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&x);
    printf("%d",findFloor(a,0,n-1,x));
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔

	Input	Expected	Got	
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.